

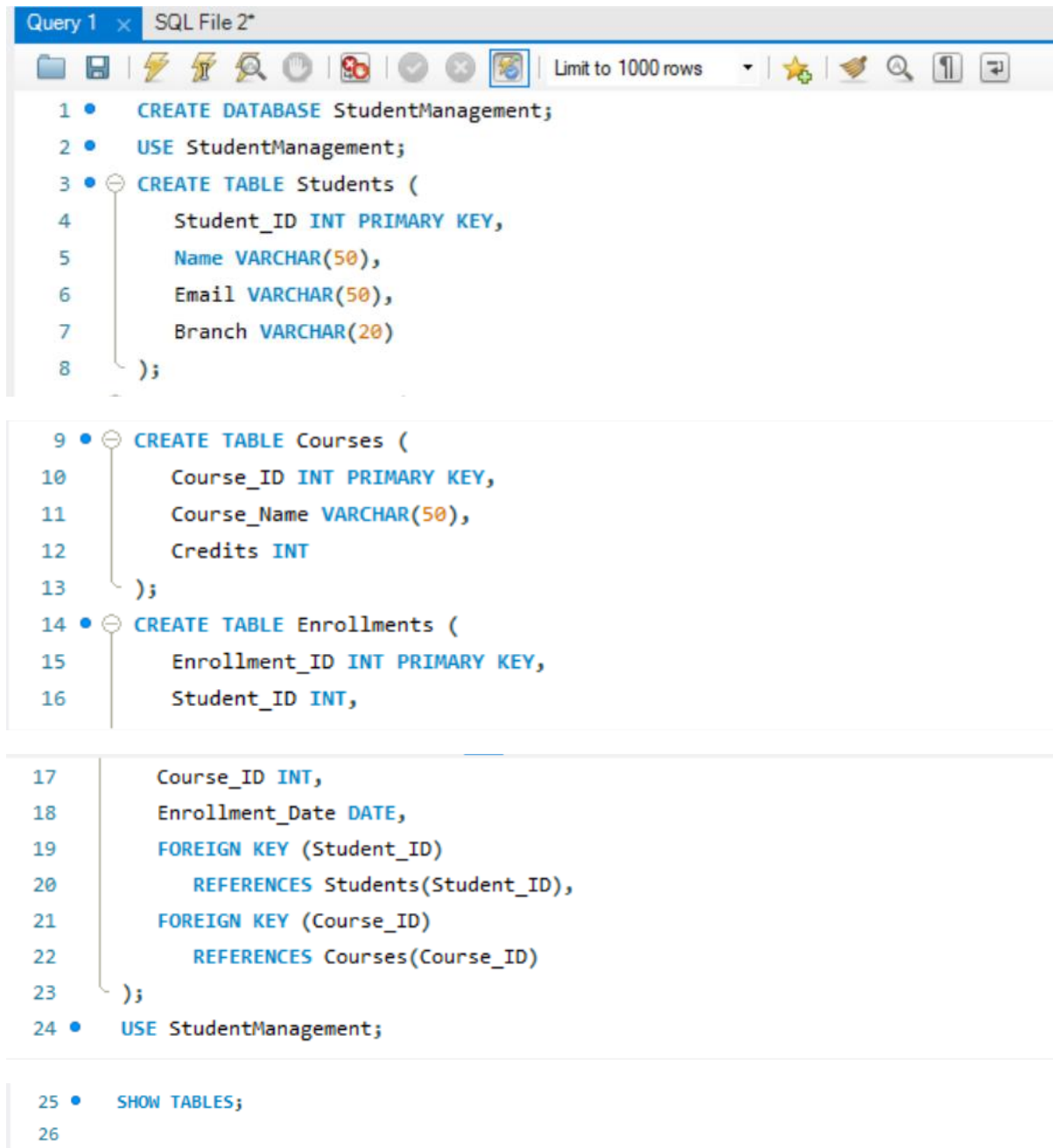
DATABASE SCHEMA INTERNSHIP TASKS

Nandini Bhimineni

Yoganandini9@gmail.com

TASK 1 & TASK 2

1. STUDENT MANAGEMENT SYSTEM DATABASE
2. SQL Table Creation



```
Query 1 x SQL File 2*
Limit to 1000 rows

1 • CREATE DATABASE StudentManagement;
2 • USE StudentManagement;
3 • CREATE TABLE Students (
4     Student_ID INT PRIMARY KEY,
5     Name VARCHAR(50),
6     Email VARCHAR(50),
7     Branch VARCHAR(20)
8 );

9 • CREATE TABLE Courses (
10     Course_ID INT PRIMARY KEY,
11     Course_Name VARCHAR(50),
12     Credits INT
13 );

14 • CREATE TABLE Enrollments (
15     Enrollment_ID INT PRIMARY KEY,
16     Student_ID INT,
17     Course_ID INT,
18     Enrollment_Date DATE,
19     FOREIGN KEY (Student_ID)
20     REFERENCES Students(Student_ID),
21     FOREIGN KEY (Course_ID)
22     REFERENCES Courses(Course_ID)
23 );
24 • USE StudentManagement;

25 • SHOW TABLES;
26
```

PRIMARY KEYS

- student_id → Students table
- course_id → Courses table
- enrollment_id → Enrollments table

FOREIGN KEYS

- student_id in Enrollments references Students
- course_id in Enrollments references Courses

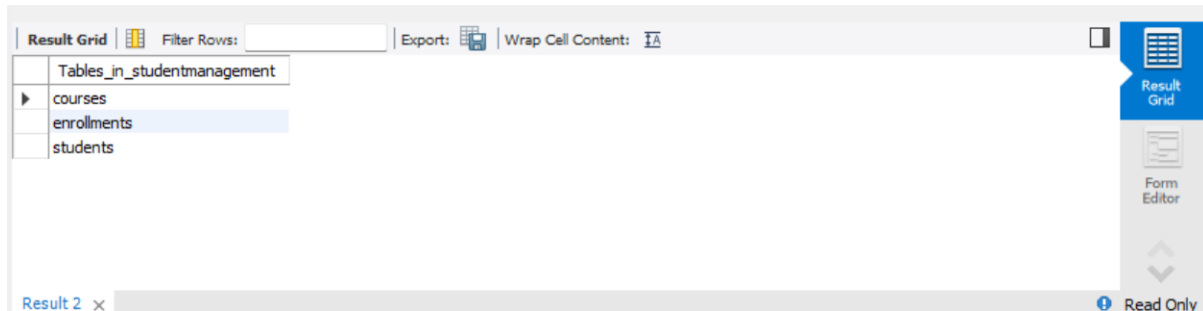
Column Name	Data Type	Purpose
Student_ID	INT	Stores unique student ID
Name	VARCHAR(50)	Stores student name
Email	VARCHAR(50)	Stores student email
Branch	VARCHAR(20)	Stores department/branch

Column Name	Data Type	Purpose
Course_ID	INT	Stores unique course ID
Course_Name	VARCHAR(50)	Stores course name
Credits	INT	Stores course credit value

Column Name	Data Type	Purpose
Enrollment_ID	INT	Stores unique enrollment ID
Student_ID	INT	Links student to enrollment
Course_ID	INT	Links course to enrollment
Enrollment_Date	DATE	Stores enrollment date

OUTPUT :

✓	12	11:13:04	USE StudentManagement	0 row(s) affected	0.000 sec
✓	13	11:13:24	SHOW TABLES	0 row(s) returned	0.032 sec / 0.000 sec
✓	14	11:14:23	USE StudentManagement	0 row(s) affected	0.000 sec
✓	15	11:14:28	CREATE TABLE Students (Student_ID INT PRIMARY KEY, Name VARCHAR...	0 row(s) affected	0.078 sec
✓	16	11:14:33	CREATE TABLE Courses (Course_ID INT PRIMARY KEY, Course_Name VAR...	0 row(s) affected	0.078 sec
✓	17	11:14:43	CREATE TABLE Enrollments (Enrollment_ID INT PRIMARY KEY, Student_ID I...	0 row(s) affected	0.093 sec
✓	18	11:14:49	USE StudentManagement	0 row(s) affected	0.000 sec
✓	19	11:14:54	SHOW TABLES	3 row(s) returned	0.016 sec / 0.000 sec



TASK 3

Database Relationships

Relationship	Relationship Type
Teacher → Students	One-to-Many (1:M)
User → Passport	One-to-One (1:1)
Students → Courses	Many-to-Many (M:N)
Customer → Orders	One-to-Many (1:M)

One-to-Many: One record can be linked to many records.

Example: One teacher teaches many students.

One-to-One: One record is linked to only one record.

Example: One user has one passport.

Many-to-Many: Many records can relate to many records.

Example: Students can enroll in many courses, and courses can have many students.

TASK 4

REAL-WORLD DATABASE SCHEMA

ONLINE SHOPPING

```

12     FOREIGN KEY (Customer_ID)
13         REFERENCES Customers(Customer_ID)
14 );
15 • CREATE TABLE Products (
16     Product_ID INT PRIMARY KEY,
17     Product_Name VARCHAR(50),
18     Price DECIMAL(10,2)
19 );
20 • CREATE TABLE Order_Items (
21     Item_ID INT PRIMARY KEY,
22     Order_ID INT,
23     Product_ID INT,
24     Quantity INT,
25     FOREIGN KEY (Order_ID)
26         REFERENCES Orders(Order_ID),
27     FOREIGN KEY (Product_ID)
28         REFERENCES Products(Product_ID)
29 );
30
31 );
32 • SHOW TABLES;

```

Query 1 SQL File 2* x

Limit to 1000 rows

```

1 • CREATE DATABASE OnlineShopping;
2 • USE OnlineShopping;
3 • CREATE TABLE Customers (
4     Customer_ID INT PRIMARY KEY,
5     Customer_Name VARCHAR(50),
6     Email VARCHAR(50)
7 );
8 • CREATE TABLE Orders (
9     Order_ID INT PRIMARY KEY,
10    Customer_ID INT,
11    Order_Date DATE,

```

OUTPUT:

#	Time	Action	Message	Duration / Fetch
✓ 20	11:20:48	CREATE DATABASE OnlineShopping	1 row(s) affected	0.031 sec
✓ 21	11:21:07	USE OnlineShopping	0 row(s) affected	0.000 sec
✓ 22	11:21:33	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Na...	0 row(s) affected	0.219 sec
✓ 23	11:22:03	CREATE TABLE Orders (Order_ID INT PRIMARY KEY, Customer_ID INT, O...	0 row(s) affected	0.094 sec
✗ 24	11:22:27	CREATE TABLE Order_Items (Item_ID INT PRIMARY KEY, Order_ID INT, P...	Error Code: 1824. Failed to open the referenced table 'products'	0.016 sec
✓ 25	11:23:33	CREATE TABLE Products (Product_ID INT PRIMARY KEY, Product_Name V...	0 row(s) affected	0.390 sec
✓ 26	11:23:38	CREATE TABLE Order_Items (Item_ID INT PRIMARY KEY, Order_ID INT, P...	0 row(s) affected	0.078 sec
✓ 27	11:24:16	SHOW TABLES	4 row(s) returned	0.000 sec / 0.000 sec

Result Grid Filter Rows: Exports Wrap Cell Content: [fA](#)

Tables_in_oneshopping
customers
order_items
orders
products

Tables Created

1. Customers
2. Orders
3. Products
4. Order_Items

Relationship	Type
Customers → Orders	One-to-Many
Orders → Order_Items	One-to-Many
Products → Order_Items	One-to-Many

Keys Used

Primary Keys

- Customer_ID
- Order_ID
- Product_ID
- Item_ID

Foreign Keys

- Customer_ID in Orders references Customers
- Order_ID in Order_Items references Orders
- Product_ID in Order_Items references Products

Appropriate Data Types

Column	Data Type	Reason
IDs	INT	Numeric identifiers
Names	VARCHAR(50)	Variable-length text
Email	VARCHAR(50)	Stores email addresses
Date	DATE	Stores order date
Price	DECIMAL(10,2)	Stores currency values

Column	Data Type	Reason
Quantity	INT	Stores item quantity

Your schema satisfies:

- Database creation
- Table creation
- Relationships
- Primary Keys
- Foreign Keys
- Proper data types